Math 107-01
Test 2A: October 9, 2013

Answer all of the following questions. You will have 50 minutes to complete the test. You may not use any outside assistance on this test. You may use a calculator, but the use of electronic equipment such as mp3 players, ipods, cell phones and any electronic devices other than your calculator during the test is prohibited. If you have time, check your answers for accuracy.

Be sure to show all work. Unsupported answers may not receive credit.

Good luck!

Name: ____

1. Let c represent the statement "It is cold outside."
   Let s represent the statement "It is snowing outside."
   Let g represent the statement "I wear gloves."
   Using these letters symbolize the following statements (3 points each):

   (a) It is cold and snowing outside.

   (b) I will wear gloves if it is cold outside.

   (c) If I do not wear gloves, then it is not cold and it is not snowing outside.

2. Let p represent the statement "It is sunny."
   Let q represent the statement "The game is at MHU."
   Let r represent the statement "I will go to the game."
   Rewrite the following statements in words (3 points each):

   (a) r→(p⋀ q)

(b) $\sim p \bigvee q$

3. Let p be the statement "I am invited." and q the statement "I will attend." Consider the statement p→q: "If I am invited, then I will attend." Symbolize the each of the following statements and then identify as the converse, inverse, contrapositive, negation, or none. (4 points each statement-2 points each answer)

Symbols	Relationship	
____________________	____________________	"If I will attend, then I am invited."
____________________	____________________	"If I will not attend, then I am not invited."
____________________	____________________	"If I am not invited, then I will not attend."
____________________	____________________	"I am not invited or I will not attend."
____________________	____________________	"I am invited and I will not attend."

Identify the negation of each statement (3 points each). Write the letter of your answer choice on the line provided:

4. You are not from North Carolina and you play soccer.

    a) You are from North Carolina or you do not play soccer.

    b) You are not from North Carolina or you play soccer.

    c) You are not from North Carolina and you play soccer.

    d) You are not from North Carolina or you do not play soccer.

    Negation: _____________

5. If you are a Business major, then you take Math 107.

    a) You are a Business major and you take Math 107.

    b) You are a Business major and you do not take Math 107.

    c) If you are not a Business major, then you do not take Math 107.

    d) If you are a Business major, then you do not take Math 107.

    Negation: _____________

6. No speaker will receive a parking ticket.
   a) All speakers will receive a parking ticket.
   b) Some speakers will not receive a parking ticket.
   c) Some speakers will receive a parking ticket.
   d) None of the above.

   Negation: ____________

7. All participants will register early.
   a) No participants will register early.
   b) Some participants will not register early.
   c) All participants will not register early.
   d) None of the above.

   Negation: ____________

8. Some participants will get a discount.
   a) Some participants will not get a discount.
   b) No participants will get a discount.
   c) All participants will get a discount.
   d) None of the above.

   Negation: ____________

9. Identify the converse, inverse, and contrapositive of the statement (3 points each);

   If I do not take Finite Mathematics, then I take Calculus.
   a) If I take Finite Mathematics, then I do not take Calculus.
   b) If I take Calculus, then I do not take Finite Mathematics.
   c) I take Calculus if I do not take Finite Mathematics.
   d) If I don't take Calculus, then I take Finite Mathematics.

   Converse: ____________

   Inverse: ____________

   Contrapositive: ____________

10. Determine whether the following are true (T) or false (F) (2 points each):

a) T$\rightarrow$F=______ b) F$\bigvee$F=______

c) F$\rightarrow$F=______ d) T$\bigwedge$T=______

e) T$\rightarrow$T=______ f) F$\bigwedge$T=______

11. (5 points) Make a truth table for the following statement: $(a \bigwedge \sim b) \rightarrow\sim c$

a	b	c				
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

12. Use a Venn diagram to determine whether the following arguments are valid or invalid (5 points each):

(a)
1. All triangles are polygons.
2. Some polygons are regular.

Therefore, all triangles are regular.

[redacted]

Valid or Invalid?____________

(b)
1. All professors wear watches
2. Kate is a professor.

Therefore, Kate wears a watch.

[redacted]

Valid or Invalid?____________

13. Use a truth table to determine whether the following arguments are valid or invalid:

(a) (6 points)

$$\begin{array}{l} 1.\ p \bigvee q \\ 2.\ \sim p \\ \hline \therefore q \end{array}$$

		1	2		c	Argument
p	q					
T	T					
T	F					
F	T					
F	F					

Valid or Invalid?__________________

(b) (8 points)

$$\begin{array}{l} 1.\ f \to \sim g \\ 2.\ g \bigvee \sim r \\ \hline \therefore f \to r \end{array}$$

					1	2		c	Argument
f	g	r							
T	T	T							
T	T	F							
T	F	T							
T	F	F							
F	T	T							
F	T	F							
F	F	T							
F	F	F							

Valid or Invalid?____________________

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University. ____________________________
(please sign)